Offline Reinforcement Learning for Sepsis Treatment COMP 579 Final Project Report (Draft)

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1 Motivation and Problem Setting

Sepsis remains a major cause of mortality in intensive care units. Clinical treatment requires sequential decisions over intravenous fluids and vasopressors under substantial uncertainty. This is naturally modeled as a finite-horizon Markov Decision Process (MDP), where states represent patient physiology, actions represent treatment decisions, and terminal outcomes reflect survival-oriented objectives. Because online experimentation is unsafe in this domain, we target *offline reinforcement learning* (offline RL): learning solely from historical logged trajectories.

Our project goal is to evaluate whether a conservative offline RL policy can outperform a behavior-cloning baseline under off-policy evaluation (OPE), while maintaining methodological discipline around data leakage, support mismatch, and reproducibility.

2 Related Work and Course Connection

Our approach is grounded in the offline RL literature and directly matches COMP 579 content on off-policy learning and policy evaluation.

- **Conservative Q-Learning (CQL)** penalizes unsupported out-of-distribution actions and is designed for stable offline value learning (Kumar et al., 2020).
- **Behavior Cloning (BC)** provides an imitation baseline and reflects the logged clinician policy.
- **Weighted Importance Sampling (WIS)** provides an off-policy value estimator with lower variance than ordinary IS in many practical settings.
- The pipeline aligns with lecture themes on off-policy RL, policy evaluation, and offline/batch RL (Feb 12, Mar 19, Mar 26 in the course schedule).

3 Data and Cohort Construction

3.1 Data Source

We implemented and validated the full pipeline on the publicly available `mimic-iv-clinical-database-demo-on-fhir-2.0` FHIR subset. This serves as an engineering and methodological proxy before running on full credentialed MIMIC-IV data.

3.2 Sepsis Cohort Logic

We select ICU encounters (`EncounterICU`) whose linked conditions include sepsis-related ICD patterns (e.g., A40/A41, 038, textual sepsis terms). Because FHIR encounter references may not always align one-to-one with ICU stay IDs, events are mapped to ICU windows by patient and timestamp when needed.

3.3 State, Action, Reward

- **State**: top 16 frequent numeric lab features from `ObservationLabevents` (code-frequency based), represented as hourly-updated vectors.
- **Action**: discretized 2D treatment index from fluid amount and vasopressor intensity bins.
- **Reward**: terminal survival proxy (+1 if alive at 30-day window, -1 if deceased), with 0 intermediate rewards.

3.4 Methodological Safeguards

- Patient-level **train/val/test** split (no transition-level mixing).
- Behavior policy probabilities estimated from **train split only** (Laplace-smoothed frequency estimator) to reduce leakage.
- Reproducible seeds and script-based execution for all stages.

4 Methods

4.1 Offline RL Algorithms

BC baseline: discrete behavior cloning with MLP encoder via `d3rlpy`.
CQL: discrete CQL with conservative penalty coefficient α and fixed optimizer/architecture defaults from `d3rlpy`.

4.2 Evaluation Protocol

We use Weighted Importance Sampling (WIS) with bootstrap confidence intervals:

$$\hat{V}_{\text{WIS}} = \frac{\sum_i w_i G_i}{\sum_i w_i}, \quad w_i = \prod_t \frac{\pi(a_t \mid s_t)}{\mu(a_t \mid s_t)}$$

where μ is the estimated behavior policy and π is the candidate policy. We evaluate on held-out split data and report 95% bootstrap CIs.

5 Implementation and Reproducibility

Pipeline stages are script-based:

- Data extraction: `src/data/extract_sepsis_cohort.py`
- Dataset export: `src/data/build_mdp_dataset.py`
- Training: `src/train/train_bc.py`, `src/train/train_cql.py`
- OPE: `src/ope/wis_eval.py`
- End-to-end runner: `experiments/run_all.py`

The project includes compatibility handling for `d3rlpy` API differences (dataset loading and constructor signatures), enabling consistent execution across local environments.

6 Extended Results and Diagnostics (Draft)

This section is intentionally expanded and detailed. We prioritize methodological transparency and diagnostic completeness over page count at this draft stage; we will compress and condense in the final submission.

6.1 Primary FHIR Demo Cohort Outcomes

Current held-out FHIR demo outputs:

- Cohort ICU encounters: 30
- Sepsis patients: 16
- Transitions: 1320
- State dimension: 16
- Split counts (train/val/test transitions): 1020 / 126 / 174

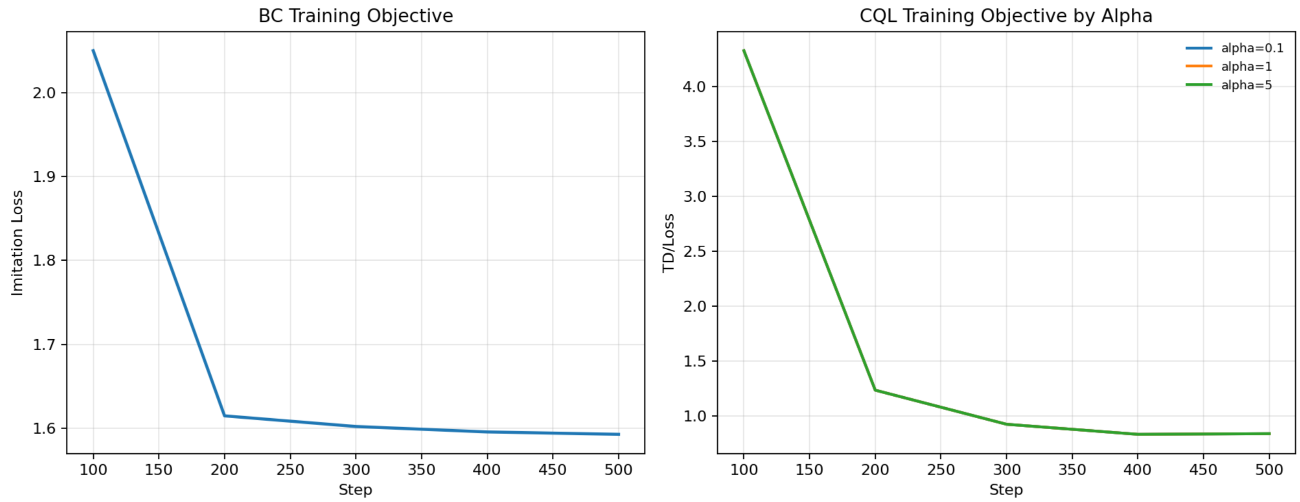

Figure 1: Training objective curves for BC and CQL variants. BC uses imitation loss; CQL panel overlays multiple α settings.

- Held-out evaluation episodes (test split): 4

WIS summary (test split, 95% CI):

Policy	Mean Return	95% CI Low	95% CI High
Behavior (logged)	-0.500	–	–
BC (μ estimated by BC model)	-0.503	-1.000	0.500
CQL ($\alpha = 1.0$)	-0.987	-1.000	-0.972

Interpretation: for this tiny held-out slice (4 test episodes), uncertainty remains high and the estimator is sensitive. The goal at this stage is primarily to validate end-to-end correctness and diagnostic behavior, not to assert definitive clinical superiority.

6.2 Training Dynamics (Figure 2 Analogue)

Figure 1 shows objective curves automatically exported from `d3rlpy` logs and collected by our report asset builder.

We observe stable BC convergence in the available runs and distinct optimization regimes across CQL conservative strengths. This is useful for diagnosing optimization stability and selecting candidate regularization ranges before full-cohort sweeps.

6.3 Feature Importance Proxy (Figure 4 Analogue)

Figure 2 shows a Random Forest feature-importance proxy computed on terminal episode outcomes from the current extracted state vectors.

Top-ranked codes in the current run include 51301, 50868, 50970, and 50912. At this stage these are engineering-level indicators that the state representation is not degenerate; final clinical interpretation will require canonical variable mapping on full MIMIC-IV.

6.4 Action Distribution Diagnostics (Figure 5 Analogue)

Figure 3 compares action-distribution mass between logged clinician actions and learned policies on evaluation states.

To quantify policy-style differences, we report Jensen-Shannon divergence:

Distribution Pair	JS Divergence
Clinician vs BC	0.208
Clinician vs CQL	0.014
BC vs CQL	0.157

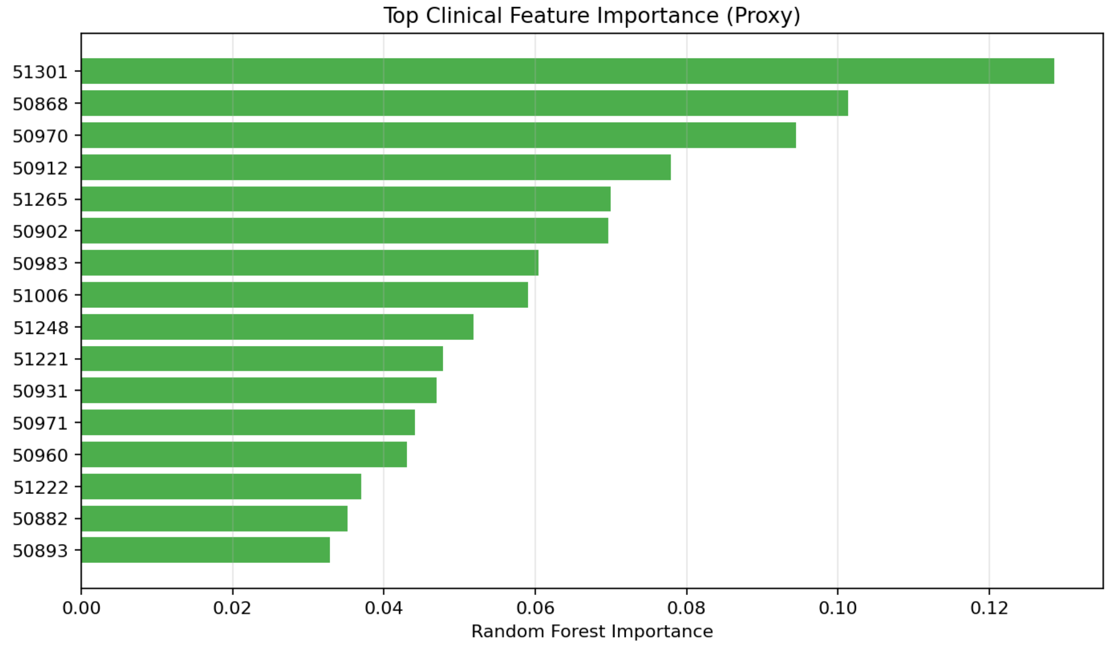

Figure 2: Top feature importance scores (proxy) from a Random Forest model over terminal outcomes. Feature labels are current lab-code identifiers from the extraction stage.

These diagnostics suggest CQL is currently closer to logged behavior than BC in this run, while BC and CQL remain meaningfully different from one another.

6.5 Value-Survival Relationship and Final Comparison (Figures 6 and 7 Analogues)

Figure 4 plots expected return against a bounded survival proxy (used here as a monotonic sanity-check transform during draft analysis).

Figure 5 summarizes final policy-level survival-proxy comparisons with CI-derived uncertainty bars where available.

Important caveat: the survival-proxy mapping used in this draft is a monotonic transformation of estimated value and therefore cannot be interpreted as calibrated clinical survival probability. It is included as a consistency diagnostic only.

6.6 Open Benchmark Validation: ICU-Sepsis

To reduce execution risk while scaling to credentialed MIMIC-IV, we maintain an open benchmark track using `Sepsis/ICU-Sepsis-v2`.

Current benchmark summary:

- Episodes: 50
- Transitions: 359
- State dimension: 716 (one-hot encoding of discrete environment state)
- Test split episodes: 8

WIS summary on ICU-Sepsis test split:

Policy	Mean Return	95% CI Low	95% CI High
Behavior (logged)	0.750	–	–
BC	0.772	0.429	1.000
CQL ($\alpha = 1.0$)	0.786	0.498	1.000

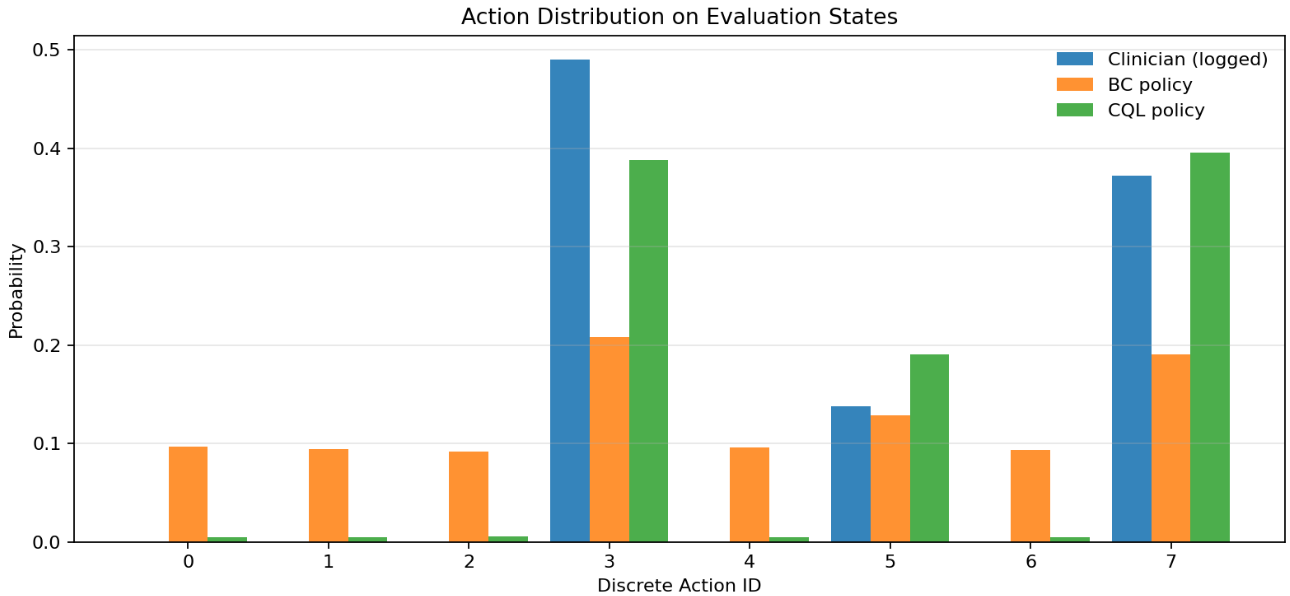

Figure 3: Action distribution comparison between logged clinician decisions and BC/CQL policy distributions.

Interpretation: this benchmark confirms algorithmic plumbing and improves confidence in reproducibility, but does not substitute for full-scale clinical validation on credentialed MIMIC-IV.

7 Limitations and Validity Threats

- The demo FHIR subset is not representative of full MIMIC-IV scale.
- OPE policy probabilities for BC/CQL are currently proxy-level approximations in this draft pipeline; final submission should replace this with stronger policy-probability estimation or additional doubly robust analyses.
- Action discretization and state selection are pragmatic first-pass choices and require sensitivity analysis.

8 Planned Final Experiments on Credentialed MIMIC-IV

1. Finalize sepsis cohort protocol and feature map with train-only normalization.
2. Run CQL hyperparameter sweeps (α and learning rate) across multiple random seeds.
3. Report subgroup and split sensitivity checks.
4. Extend OPE robustness (clipping sweeps, variance analysis, optional doubly robust estimator).

9 Contributions

- **Alex Groiser**: data engineering, FHIR cohort extraction, MDP construction, reproducibility scaffolding.
- **Yusuf Hamza**: training/evaluation integration, OPE pipeline, experiment orchestration and analysis.
- **Shared**: report writing, interpretation, final validation and presentation.

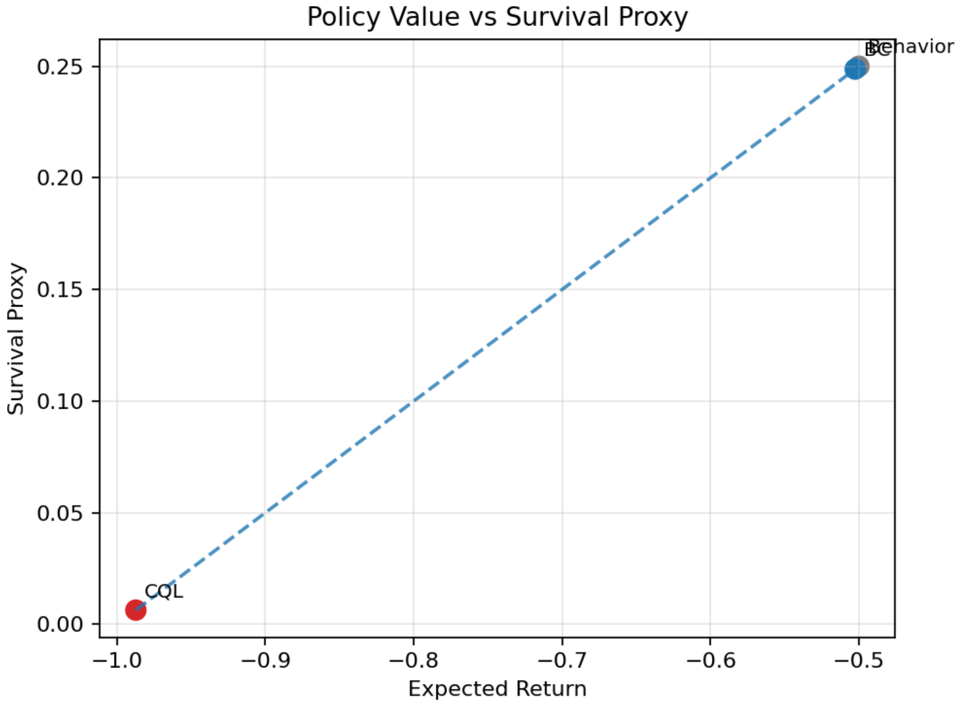

Figure 4: Relationship between policy expected return and survival proxy.

References

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3. Johnson, A. E. W., et al. (2023). *MIMIC-IV, a freely accessible electronic health record dataset.* Scientific Data.

4. Levine, S., Kumar, A., Tucker, G., and Fu, J. (2020). *Offline Reinforcement Learning: Tutorial, Review, and Perspectives on Open Problems.* arXiv:2005.01643.

5. Wang, Z., et al. (2023). *Diffusion Policies as an Expressive Representation for Offline RL.* ICLR.

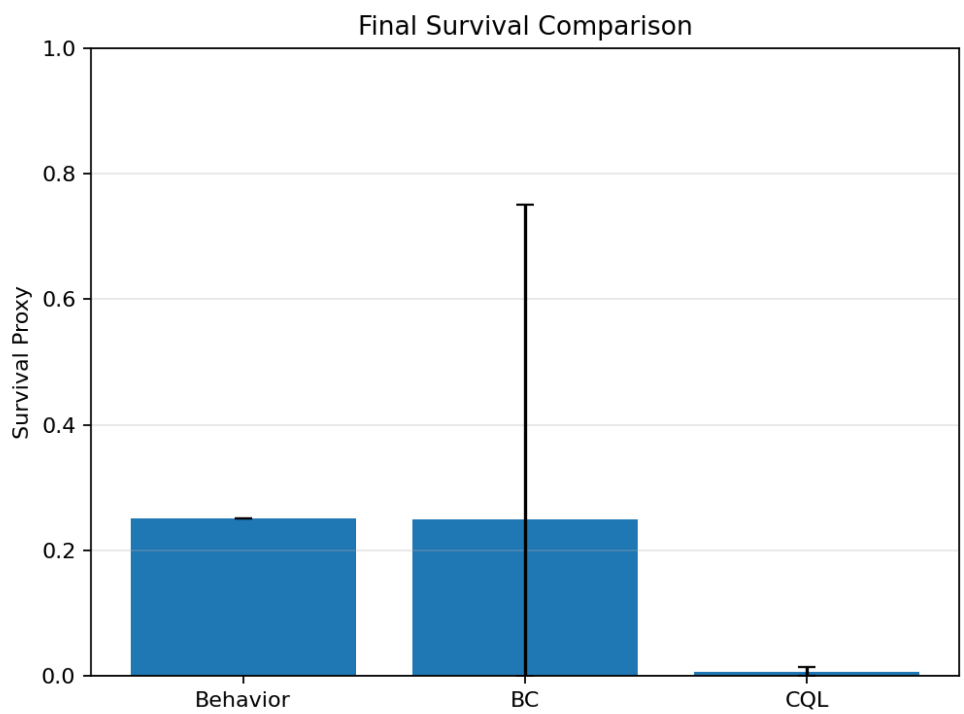

Figure 5: Final policy comparison using survival-proxy transformation from expected returns.